Practical No. 2 : Alpha-Beta Pruning

```
import math

values = [3, 5, 6, 9, 1, 2, 0, -1]

def alphabeta(d, n, is_max, alpha, beta):
    if d == 3:
        return values[n]

    if is_max:
        best = -math.inf
        for i in range(2):
            best = max(best, alphabeta(d+1, n*2+i, False, alpha, beta))
            alpha = max(alpha, best)

            if beta <= alpha:
                print("Pruning at MAX node")
                break
        return best

    else:
        best = math.inf
        for i in range(2):
            best = min(best, alphabeta(d+1, n*2+i, True, alpha, beta))
            beta = min(beta, best)

            if beta <= alpha:
                print("Pruning at MIN node")
```

```
            break
        return best

print("Leaf Node Values:", values)

result = alphabeta(0, 0, True, -math.inf, math.inf)

print("\nOptimal Value:", result)
print("\nLogic:")
print("Alpha stores the best value found by MAX.")
print("Beta stores the best value found by MIN.")
print("When alpha >= beta, remaining branches are pruned.")
print("This reduces the number of nodes evaluated.")
```